



Brief paper

On the structural output controllability and functional observability of undirected networks[☆]Yuan Zhang, Ranbo Cheng^{*}, Yuanqing Xia

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ABSTRACT

In this paper, we investigate the structural output controllability (SOC) and structural functional observability (SFO) of undirected networks. We first characterize the generic dimensions of controllable subspace and output controllable subspace of undirected networks, relating them to either the generic ranks of simple products of matrices without the symmetric weight constraint, or maximum linkings of certain dynamic graphs. Our results show that, with or without the symmetric weight constraint, these dimensions always coincide. Our characterizations also induce a maximum flow based algorithm for computing the generic dimension of output controllable subspace in $O(n^{2.5})$ time (n is the system state dimension). This challenges a claim of its NP-hardness in a recent work. Additionally, we demonstrate that minimizing the number of driver nodes required to achieve SOC of undirected networks can be solved in $O(n^{2.5})$ time. We also show that this problem becomes NP-hard if the driver nodes need to be dedicated. Building on these results, we further develop criteria for the SFO of undirected networks, which are not directly dual to the SOC. Our approach is purely graph-theoretic, based on the dynamic graphs of undirected networks, not relying on the eigenspace of symmetric matrices. As a by-product, we also present alternative proofs for the criteria of structural (output) controllability of undirected networks.

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1. Introduction

Consider a linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t), y(t) = Cx(t), \quad (1)$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^m$, $y(t) \in \mathbb{R}^p$ are state, input, and output vectors at time t , respectively. Accordingly, $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times m}$, and $C \in \mathbb{R}^{p \times n}$. System (1) is referred to as system (A, B, C) .

The *output controllable subspace* is the subspace of $\mathbb{R}^p$ consisting of those $y(T) = Cx(T) \in \mathbb{R}^p$ such that $x(T)$ can be steered to from the initial state $x(0) = 0$ by suitable input $u(t) : [0, T] \rightarrow \mathbb{R}^m$ for some finite $T > 0$. Denote by $d(A, B, C)$ the dimension of the output controllable subspace of the system (A, B, C) . It is known that (Chen, 1984)

$$d(A, B, C) = \text{rank } C[B, AB, \dots, A^{n-1}B].$$

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When $C = I_n$, with I_n being the $n \times n$ identity matrix, the output controllable subspace reduces to the controllable subspace. We denote the dimension of controllable subspace by $d(A, B)$, i.e.,

$$d(A, B) = \text{rank } [B, AB, \dots, A^{n-1}B]$$

If $d(A, B, C) = p$, then system (A, B, C) is said to be *output controllable*.

In practical systems, especially large-scale network systems, obtaining the exact values for the parameters of (A, B, C) can be challenging. Instead, it is often easier to determine the zero-nonzero patterns of A, B and C , which indicate the connections among state variables, inputs, and outputs (Liu et al., 2011). In such scenarios, we may focus on the generic properties, which are properties that hold for almost all parameter values in the corresponding parameter space. The study of generic properties of network systems described by structured systems has received increasing attention in the past few decades; see Ramos et al. (2022). To be specific, a system is considered structured if all the system matrices are structured (Reinschke, 1988). A structured matrix is defined as a matrix with entries that are either fixed at zero or free to take arbitrary real values independently of one another (the latter class of entries is called nonzero entries). A numerical matrix $\tilde{M}$ is said to be structurally equivalent to a (structured) matrix M if $M_{ij} = 0$ implies $\tilde{M}_{ij} = 0$. Notice that if $M_{ij} \neq 0$ (i.e., M_{ij} is a nonzero entry), then $\tilde{M}_{ij}$ can take either

zero or nonzero real values. The matrix $\tilde{M}$ is called a realization of M . Let $\text{re}(A)$ ($\text{re}(A, B)$, $\text{re}(A, B, C)$, respectively) be the set of realizations of A (resp. (A, B) , (A, B, C)). We define the generic dimension of the controllable subspace of (A, B) as the maximum dimension of the controllable subspace of $(\tilde{A}, \tilde{B})$ over all realizations of (A, B) , i.e., $gd(A, B) = \max_{(\tilde{A}, \tilde{B}) \in \text{re}(A, B)} d(\tilde{A}, \tilde{B})$. Similarly, the generic dimension of the output controllable subspace of (A, B, C) is defined as $gd_s(A, B, C) = \max_{(\tilde{A}, \tilde{B}, \tilde{C}) \in \text{re}(A, B, C)} d(\tilde{A}, \tilde{B}, \tilde{C})$. It turns out that for almost all $(\tilde{A}, \tilde{B}, \tilde{C}) \in \text{re}(A, B, C)$, $d(\tilde{A}, \tilde{B}, \tilde{C}) = gd_s(A, B, C)$ (Dion et al., 2003). If $gd(A, B) = n$, (A, B) is *structurally controllable*. If $gd_s(A, B, C) = p$, (A, B, C) is called *structurally output controllable* (SOC).

In this paper, we are interested in the case where A is structurally symmetric, meaning that $A_{ij} \neq 0 \Leftrightarrow A_{ji} \neq 0$. This case arises when the interactions among state variables are bidirectional (Menara et al., 2019). A matrix $\tilde{A}$ is *symmetrically structurally equivalent* to A , if $\tilde{A}$ is structurally equivalent to A and $\tilde{A} = \tilde{A}^T$ ($(\cdot)^T$ takes the transpose). We call $\tilde{A} = \tilde{A}^T$ the *symmetric weight constraint* or *symmetry constraint* for short. Let $\text{res}(A, B)$ be the set of matrices $(\tilde{A}, \tilde{B})$ such that $\tilde{A}$ is symmetrically structurally equivalent to A and $\tilde{B}$ is structurally equivalent to B ($\text{res}(A, B, C)$ is defined similarly). The generic dimensions of the controllable subspace and output controllable subspace of (A, B, C) with the symmetry constraint are defined as:

$$gd_s(A, B) = \max_{(\tilde{A}, \tilde{B}) \in \text{res}(A, B)} d(\tilde{A}, \tilde{B}), \text{ and}$$

$$gd_s(A, B, C) = \max_{(\tilde{A}, \tilde{B}, \tilde{C}) \in \text{res}(A, B, C)} d(\tilde{A}, \tilde{B}, \tilde{C}).$$

Notice that in defining $gd(A, B)$ ($gd_s(A, B, C)$), we do not require $\tilde{A} = \tilde{A}^T$. In this paper, we aim to characterize $gd_s(A, B)$ and $gd_s(A, B, C)$, which can, in turn, characterize $gd(A, B, C)$ and structural functional observability (SFO, defined in Section 5) of undirected networks.

Related references: The concept of structural controllability has garnered significant attention since its initiation (Lin, 1974). Structural controllability captures the generic property of controllability that is mainly dominated by the zero-nonzero structure of system matrices, eliminating the need for precise values of system parameters. This notable feature enables intuitive graph-theoretic characterizations and efficient combinatorial algorithms for the system analysis and optimization related to controllability (Ramos et al., 2022), especially for large-scale networks (Liu et al., 2011).

Yet, most existing works on structural controllability assume that all nonzero (indeterminate) entries in the system matrices can take values independently. It was very recently that structural controllability and strong structural controllability (a more stringent notion than structural controllability) of undirected networks, i.e., structured systems with the symmetry constraint, have been investigated in Mousavi et al. (2017) and Menara et al. (2019). In particular, a symmetric cactus based criterion for structural controllability of undirected networks was derived in Menara et al. (2019). Later, Li et al. (2021) studied the structural target controllability and SOC of undirected networks.¹ However, both works focus on qualitative characterizations (i.e., whether a system is structurally controllable or SOC), leaving implicit the quantitative computation of $gd_s(A, B)$ and $gd_s(A, B, C)$. In fact, both works derived their results starting from the PBH test and using an algebraic-graphic manner, which combines the eigenspaces of certain matrices with related graph-theoretic concepts. It would turn out in this paper that the quantitative $gd_s(A, B, C)$ and $gd_s(A, B)$ are helpful in determining the minimal driver nodes

needed for SOC and obtaining criteria for SFO of undirected networks (see Theorems 3 and 5).

For directed networks without symmetry constraints, $gd(A, B)$ was completely characterized using the cactus cover in Hosoe (1980) and Poljak (1990). Whereas, it is still open how to compute $gd(A, B, C)$ deterministically or characterize the SOC of (A, B, C) (Murota & Poljak, 1990). Correspondingly, the problem of determining the minimum number of driver nodes (independent inputs) needed for structural controllability can be solved by the maximum matching based approach (Liu et al., 2011), while the similar problem for SOC remains open (Gao et al., 2014). It is worth noting that results on $gd(A, B, C)$ cannot be directly applied to $gd_s(A, B, C)$, since symmetric realizations of A correspond to a set of zero measure in all realizations of A .

The concept of functional observability captures the ability to uniquely determine a linear combination (function) of system initial states from the system outputs and inputs (Fernando et al., 2010), which generalizes the state observability. Although parallel to output controllability, there is no direct duality between them (Zhang et al., 2024a). Functional observability is the existence condition for a functional observer with arbitrary poles to estimate linear functions of system states (Fernando et al., 2010; Jennings et al., 2011). This concept has found applications in various fields (Montanari et al., 2022; Zhang et al., 2023, 2024a). SFO captures the generic behavior of functional observability when only the zero-nonzero patterns of system matrices are available, which has attracted much attention recently (Montanari et al., 2022; Zhang et al., 2024a). Yet, existing criteria for SFO all apply to directed networks, leaving undirected networks with symmetric weights unexplored.

Contributions: In this paper, we provide characterizations for $gd_s(A, B)$ and $gd_s(A, B, C)$ using a pure graph-theoretic approach. Based on the results for $gd_s(A, B)$, we also characterize the SFO of undirected networks. Our main contributions can be summarized as follows:

- Firstly, we present characterizations for $gd_s(A, B, C)$ in terms of both maximum linkings in certain dynamic graphs and generic ranks of simple products of matrices without the symmetry constraint. Our results show that for undirected networks, $gd(A, B, C)$ and $gd_s(A, B, C)$ always coincide, and the maximum linking based upper bound and the cactus cover based lower bound for $gd(A, B, C)$ given in Murota and Poljak (1990) also coincide. By letting $C = I_n$, these results apply to $gd_s(A, B)$ and $gd(A, B)$ as well.
- Secondly, we show that $gd_s(A, B, C)$ can be computed in $O(n^{2.5})$ time via the maximum flow algorithm, where n is the system state dimension. Consequently, verifying the SOC of undirected networks is polynomially solvable irrespective of the symmetry constraint. This corrects a claim in a recent work (Li et al., 2021) on its NP-hardness. Based on this result, we further show that the minimal driver nodes needed for SOC of undirected networks can be determined in $O(n^{2.5})$ time, a problem still open for directed networks (Gao et al., 2014). We also demonstrate that this problem becomes NP-hard if each driver node can only actuate one state variable.
- Thirdly, based on the characterization of $gd_s(A, B)$, we provide criteria for SFO of undirected networks. These criteria are not directly dual to the criteria for SOC, and can be verified in polynomial time.

Technically, our approach is based on the so-called dynamic graphs (Murota & Poljak, 1990; Poljak, 1990), without relying on the eigenspace of symmetric matrices as in Menara et al. (2019) and Li et al. (2021), making it purely graph-theoretic. As

¹ When each row of C has at most one nonzero entry, SOC is also called structural target controllability, in which the nonzero columns of C correspond to the target nodes to be controlled.

a by-product, we also offer an alternative proof for the characterizations of structural (output) controllability of undirected networks given in Menara et al. (2019) and Li et al. (2021). This paper is organized as follows. Section 2 reviews basic preliminaries. Section 3 characterizes $gd_s(A, B, C)$ and clarifies its computational complexity. Section 4 investigates the problem of minimizing the number of driver nodes required for SOC of undirected networks. Section 5 characterizes the SFO of undirected networks. The last section concludes this paper.

2. Preliminaries

We denote a directed graph (digraph) by $\mathcal{G} = (V, E)$, where V is the vertex set, $E \subseteq V \times V$ is the edge set. We say $\mathcal{G}$ covers V_s if $V_s \subseteq V$. A subgraph of $\mathcal{G}$ is a digraph $\mathcal{G}_s = (V_s, E_s)$ such that $V_s \subseteq V$, $E_s \subseteq (V_s \times V_s) \cap E$. In addition, $\mathcal{G}_s$ is induced by V_s if $E_s = (V_s \times V_s) \cap E$. A path P is a sequence of edges such that the initial vertex of each edge is the terminal vertex of its preceding one. The length of a path P , denoted by $|P|$, is the number of edges it contains. A path forms a cycle when its start and end vertices are identical, and it does not contain any other repeated vertices along its route. A cycle with only one vertex is called a self-loop. If every edge of $\mathcal{G}$ is bidirectional (i.e., $(v_i, v_j) \in E \Leftrightarrow (v_j, v_i) \in E$), or simply put, $\mathcal{G}$ is undirected, a connected component of $\mathcal{G}$ is a maximal connected subgraph, meaning it is not possible to add any more edges or vertices to it from $\mathcal{G}$ without breaking the property of being connected (i.e., any two vertices are connected to each other by paths).

Associated with (A, B, C) , define digraph $\mathcal{G}(A, B, C) = (X \cup U \cup Y, E_{XX} \cup E_{UX} \cup E_{XY})$, with state vertices $X = \{x_1, \dots, x_n\}$, input vertices $U = \{u_1, \dots, u_m\}$, output vertices $Y = \{y_1, \dots, y_p\}$, edges $E_{XX} = \{(x_j, x_i) : A_{ij} \neq 0\}$, $E_{UX} = \{(u_j, x_i) : B_{ij} \neq 0\}$, and $E_{XY} = \{(x_i, y_j) : C_{ji} \neq 0\}$. A set $X_s \subseteq X$ can be *matched into* Y in $\mathcal{G}(A, B, C)$ if there exist $|X_s|$ vertex-disjoint edges from X_s to Y in $\mathcal{G}(A, B, C)$. Building on $\mathcal{G}(A, B, C)$, we define $\mathcal{G}(A) = (X, E_{XX})$, $\mathcal{G}(A, B) = (X \cup U, E_{XX} \cup E_{UX})$, and $\mathcal{G}(A, C) = (X \cup Y, E_{XX} \cup E_{XY})$. In other words, $\mathcal{G}(A)$, $\mathcal{G}(A, B)$, and $\mathcal{G}(A, C)$ are subgraphs of $\mathcal{G}(A, B, C)$ induced by X , $X \cup U$, and $X \cup Y$, respectively. A *stem* is a path with no repeated vertices from an input vertex to a state vertex in $\mathcal{G}(A, B)$. A state vertex $x_i \in X$ is said to be *input-reachable*, if it is the terminal vertex of a stem. A set $X_s \subseteq X$ is said to be *covered by a cactus configuration* in $\mathcal{G}(A, B)$ if the following conditions hold: (1) every $x_i \in X_s$ is input-reachable, and (2) X_s can be covered by a collection of vertex-disjoint cycles and stems. Notice that any subset of the covered set X_s is also covered.

A bipartite graph $\mathcal{B} = (V, E)$, also denoted by $\mathcal{B} = (V_L, V_R, E)$, is a graph whose vertex set V can be partitioned into two disjoint subsets V_L and V_R (called the components of V), such that $E \subseteq V_R \times V_L$.² A matching of $\mathcal{B}$ is a set of edges $\mathcal{M} \subseteq E$ in which no two edges share a common vertex (including initial and terminal vertices). The size of $\mathcal{M}$, given by $|\mathcal{M}|$, is the number of edges in $\mathcal{M}$. A vertex $v \in V_L$ (resp. $v \in V_R$) is left (resp. right) matched by $\mathcal{M}$ if it is incident to an edge in $\mathcal{M}$. A set $V_s \subseteq V_L$ is (left) matched by a matching $\mathcal{M}$, if every vertex in V_s is (left) matched by $\mathcal{M}$. A maximum matching of $\mathcal{B}$ is a matching with the largest size among all matchings of $\mathcal{B}$. Associated with (A, B) , define $\mathcal{B}(A, B) = (X, X \cup U, E_{XX} \cup E_{UX})$ and $\mathcal{B}(A) = (X, X, E_{XX})$, in which with a little abuse of terminology, we use the same labels for vertices in different components to present the one-one correspondences between edges of $\mathcal{B}(A, B)$ and edges of $\mathcal{G}(A, B)$.

By $\text{grank } M$, we mean the maximum rank that a realization of a structured matrix M can take *without the symmetry constraint*,

² Although edges in bipartite graphs are undirected, in this paper it is convenient to consider that they have a direction from the right component V_R to the left component V_L .

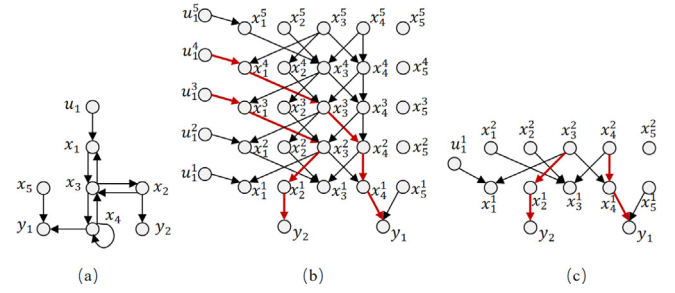


Fig. 1. Subfigure (a) is the $\mathcal{G}(A, B, C)$ of a system (A, B, C) , (b) is the dynamic graph $\mathcal{D}(A, B, C)$, and (c) is the corresponding $\mathcal{D}_r(A_r, B, C)$. In (b), bold (red) edges form a linking L of size 2 from V_B to V_C , and its corresponding linking L' of size 2 from $V_B^{(1)} \cup V_A^{(2)}$ to V_C in $\mathcal{D}_r(A_r, B, C)$ is also depicted with bold (red) lines in (c). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

i.e., $\text{grank } M = \max_{\tilde{M} \in \text{res}(M)} \text{rank } \tilde{M}$. It is known that $\text{grank } M$ equals the maximum size of a matching in $\mathcal{B}(M)$ (Murota, 2009), where the bipartite graph $\mathcal{B}(M) = (V_L, V_R, E)$, in which V_L and V_R correspond to the rows and columns of M , respectively, and $E = \{(j, i) : M_{ij} \neq 0\}$. For a sequence of matrices $M_1, \dots, M_r$ with compatible dimensions, $\text{grank } \prod_{i=1}^r M_i$ denotes the maximum rank the product $\prod_{i=1}^r \tilde{M}_i$ can take, where $\tilde{M}_i$ is a realization of M_i for each i . If M is structurally symmetric, $\text{grank}_s M$ denotes the maximum rank of matrices that are symmetrically structurally equivalent to M , i.e., $\text{grank}_s M = \max_{\tilde{M} \in \text{res}(M)} \text{rank } \tilde{M}$.

3. Characterizations of $gd_s(A, B, C)$

In this section, we characterize $gd_s(A, B, C)$ based on the so-called dynamic graphs (Murota & Poljak, 1990).

Associated with (A, B, C) , the dynamic graph $\mathcal{D}(A, B, C)$ is defined on the vertex set $V_A \cup V_B \cup V_C$ with $V_A = \{x_i^k : i = 1, \dots, n, k = 1, \dots, n\}$, $V_B = \{u_i^k : i = 1, \dots, m, k = 1, \dots, n\}$, and $V_C = \{y_1, \dots, y_p\}$, and the edge set $\{(x_j^{k+1}, x_i^k) : A_{ij} \neq 0, k = 1, \dots, n-1\} \cup \{(u_j^k, x_i^k) : B_{ij} \neq 0, k = 1, \dots, n\} \cup \{(x_i^k, y_j) : C_{ji} \neq 0\}$. Let $V_A^{(k)} = \{x_i^k : i = 1, \dots, n\}$ and $V_B^{(k)} = \{u_i^k : i = 1, \dots, m\}$ for $k = 1, \dots, n$. Here, x_i^k (u_i^k , respectively) is the k th copy of the state vertex x_i (input vertex u_i), $k = 1, \dots, n$. Upon letting $A = [a_{ij}]$, $B = [b_{ij}]$, and $C = [c_{ij}]$, the weight of an edge in $\mathcal{D}(A, B, C)$ is set as

$$w(e) = \begin{cases} a_{ij}, & \text{if } e = (x_j^{k+1}, x_i^k), k = 1, \dots, n-1 \\ b_{ij}, & \text{if } e = (u_j^k, x_i^k), k = 1, \dots, n \\ c_{ij}, & \text{if } e = (x_i^k, y_j). \end{cases}$$

Fig. 1(b) provides an illustration of a dynamic graph $\mathcal{D}(A, B, C)$ whose associated digraph $\mathcal{G}(A, B, C)$ is given in Fig. 1(a).

A collection $L = (p_1, \dots, p_k)$ of vertex-disjoint paths in $\mathcal{D}(A, B, C)$ is called a *linking*, and the size of a linking is the number k of paths it contains. See Fig. 1(b) for the illustration of a linking with size 2. Let $\text{tail}(L)$ and $\text{head}(L)$ be respectively the set of initial vertices and the set of terminal vertices of paths in L . A linking L is called a (S, T) linking, if $S = \text{tail}(L)$, and $T = \text{head}(L)$. For a linking $L = (p_1, \dots, p_k)$, its weight $w(L)$ is defined as the product of the weights of all individual edges of the paths p_i in L , $i = 1, \dots, k$. Let $<$ be a fixed order of vertices in $\mathcal{D}(A, B, C)$. We are interested in the linkings from V_B to V_C of $\mathcal{D}(A, B, C)$, which are (J, I) linkings with $J \subseteq V_B$ and $I \subseteq V_C$. For such a linking $L = (p_1, \dots, p_k)$, suppose $\text{tail}(L) = \{s_1, \dots, s_k\}$ and $\text{head}(L) = \{t_1, \dots, t_k\}$ such that $s_1 < \dots < s_k$ and $t_1 < \dots < t_k$. Moreover, suppose that $s_{\pi(i)}$ and t_i are respectively the initial vertex and terminal vertex of p_i , $\pi(i) \in \{1, \dots, k\}$,

$i = 1, \dots, k$. Then, $\pi = (\pi(1), \dots, \pi(k))$ is a permutation of $(1, \dots, k)$. The sign of L is defined as $\text{sign}(L) \doteq \text{sign}(\pi)$, where $\text{sign}(\pi)$ is the sign of permutation π . Recall that $\text{sign}(\pi) = (-1)^q$, where q is the number of transpositions (exchanges between two positions) required to transform the permutation π into an ordered sequence.

Define matrix

$$W \doteq [CB, CAB, \dots, CA^{n-1}B]. \quad (2)$$

Observe that there are one-one correspondences between rows of W and the set V_C , and the columns of W and the set V_B . Let $W(I, J)$ be the submatrix of W consisting of rows indexed by I and columns by J , $I \subseteq V_C, J \subseteq V_B$. The following lemma relates the determinant of a square submatrix of W to linkings in $\mathcal{D}(A, B, C)$, which is the base of the subsequent derivations. It is noted that (3) in this lemma could also be obtained from the determinant expressions in Reinschke (1988, Sec A2), independent of the dynamic graph representations.

Lemma 1 (Murota and Poljak, 1990). *Let $W(I, J)$ be a square submatrix of W . Then,*

$$\det W(I, J) = \sum_L \text{sign}(L)w(L), \quad (3)$$

where the summation is taken over all (J, I) linkings from V_B to V_C of $\mathcal{D}(A, B, C)$, and J and I are regarded as subsets of V_B and V_C , respectively.

Let Q be the controllability matrix of (A, B) , i.e.,

$$Q \doteq [B, AB, \dots, A^{n-1}B]. \quad (4)$$

Let $\mathcal{D}(A, B)$ be the subgraph of $\mathcal{D}(A, B, C)$ induced by $V_A \cup V_B$. Similarly, the linkings in $\mathcal{D}(A, B)$ from V_B to $V_A^{(1)}$ can be used to characterize Q in the same way as (3) (Poljak, 1990). Furthermore, let $X_{nr} \subseteq X$ be the set of input-unreachable vertices in $\mathcal{G}(A, B)$. Let A_r be obtained from A by zeroing the rows and columns corresponding to X_{nr} . Define a dynamic graph $\mathcal{D}_r(A_r, B, C)$ with vertex set $V_A^2 \cup V_B^{(1)} \cup V_C$, where $V_A^2 = V_A^{(2)} \cup V_A^{(1)}$, and $V_A^{(i)}, V_B^{(i)}$ and V_C are defined as previously, and the edge set $\{(x_i^2, x_i^1) : [A_r]_{ij} \neq 0\} \cup \{(u_j^1, x_i^1) : B_{ij} \neq 0\} \cup \{(x_i^1, y_j) : C_{ji} \neq 0\}$. See Fig. 1(c) for the illustration of $\mathcal{D}_r(A_r, B, C)$. Denote $\mu(A, B, C)$ as the maximum size of a linking from V_B to V_C in $\mathcal{D}(A, B, C)$, and let $\mu_r(A_r, B, C)$ be the maximum size of a linking from $V_B^{(1)} \cup V_A^{(2)}$ to V_C in $\mathcal{D}_r(A_r, B, C)$. It turns out that $\mathcal{D}_r(A_r, B, C)$ is the subgraph of $\mathcal{D}(A_r, B, C)$ induced by $V_A^{(2)} \cup V_A^{(1)} \cup V_B^{(1)} \cup V_C$. Thus, for any linking L of size k from V_B to V_C in $\mathcal{D}(A, B, C)$, we can always obtain a linking L' of size k in $\mathcal{D}_r(A_r, B, C)$ by taking the sub-paths of L from $V_B^{(1)} \cup V_A^{(2)}$ to V_C ; see Fig. 1 for illustration (noting that any state vertex in a path of L corresponds to an input-reachable state vertex in $\mathcal{G}(A, B)$). With this observation, we summarize some intuitive relationships among $gd_s(A, B, C)$, $\mu(A, B, C)$, and $\mu_r(A_r, B, C)$ as follows.

Lemma 2. *For a structurally symmetric A , it holds*

$$gd_s(A, B, C) \leq gd(A, B, C) \leq \mu(A, B, C) \leq \mu_r(A_r, B, C).$$

Proof. It is apparent that $gd_s(A, B, C) \leq gd(A, B, C)$, since the symmetry constraint is removed in $gd(A, B, C)$. If $gd(A, B, C) = k \geq 1$, then there exists at least one non-singular $k \times k$ square submatrix in W . From Lemma 1, the maximum size of a linking from V_B to V_C in $\mathcal{D}(A, B, C)$ is at least k . This leads to $gd(A, B, C) \leq \mu(A, B, C)$. Now suppose $L = (V_L, E_L)$ is a linking from V_B to V_C with the maximum size in $\mathcal{D}(A, B, C)$. Suppose further $V_L \cap V_A^{(j)} = \tilde{V}_A^{(j)}$, $j = 1, 2$, $V_L \cap V_B^{(1)} = \tilde{V}_B^{(1)}$, and $V_L \cap V_C = \tilde{V}_C$. Notice that every vertex $x_i^1 \in \tilde{V}_A^{(1)}$ is the terminal vertex of a path from some input vertex from V_B in $\mathcal{D}(A, B, C)$, leading to that x_i is input-reachable

in $\mathcal{G}(A, B)$. From this fact, it is easy to see that the subgraph of L induced by $\tilde{V}_A^{(1)} \cup \tilde{V}_A^{(2)} \cup \tilde{V}_B^{(1)} \cup \tilde{V}_C$ is a linking from $V_B^{(1)} \cup V_A^{(2)}$ to V_C in $\mathcal{D}_r(A_r, B, C)$. As a result, $\mu(A, B, C) \leq \mu_r(A_r, B, C)$. $\square$

From the proof above, it is clear that $gd(A, B, C) \leq \mu(A, B, C) \leq \mu_r(A_r, B, C)$ holds even when A is not structurally symmetric. In what follows, we will show that all three inequalities in Lemma 2 can indeed be replaced with equalities by delving into the symmetric structure of A . To this end, we first introduce some intermediate results, including Lemmas 3–6 and Propositions 1 and 2. In particular, Lemma 3 describes a decomposition of $\mathcal{G}(A)$ according to maximum matchings of $\mathcal{B}(A)$, which is valid regardless of whether A is structurally symmetric or not. Lemma 4 extends this result to the bipartite graph $\mathcal{B}(A, B)$ with A being structurally symmetric. Afterwards, Proposition 1 links the condition $\mu_r(A_r, B, C) = k$ to a cactus cover condition, based on Lemmas 5 and 6. Proposition 2 then establishes the relation between the cactus cover condition and the full rank of subrows of the controllability matrix Q . All these results together lead to the subsequent Theorem 1.

Lemma 3. (Pequito et al., 2016, Lem 3) *Suppose $\mathcal{M}$ is a maximum matching of $\mathcal{B}(A)$. Then, $(X, \mathcal{M})$ is a union of disjoint cycles and paths that spans $\mathcal{G}(A)$, where the number of paths equals $n - |\mathcal{M}|$ (an isolated vertex is regarded as a path with length zero).*

Lemma 4. *Suppose A is structurally symmetric. Let $\mathcal{M}$ be a maximum matching of $\mathcal{B}(A, B)$. Then, $(X \cup U, \mathcal{M})$ is a union of disjoint cycles (consisting of state vertices, called state cycles) and paths that spans $\mathcal{G}(A, B)$, where the number of paths equals $n + m - |\mathcal{M}|$, and if a path starts from a state vertex, then its length must be even.*

Proof. Let us consider the matrix $F = \begin{bmatrix} A & B \\ 0_{m \times n} & 0_{m \times m} \end{bmatrix}$. It is clear that $\text{grank } F = \text{grank } [A, B] = |\mathcal{M}|$. The utilization of Lemma 3 on $\mathcal{B}(F)$ yields that $(X \cup U, \mathcal{M})$ is a union of disjoint state cycles and paths that spans $\mathcal{G}(A, B)$, noting that there exists no edge from X to U . Moreover, the number of paths is $n + m - |\mathcal{M}|$. Assume that there is a path $P = (x_{i_1}, x_{i_2}, \dots, x_{i_k})$ with an odd length, i.e., $k \geq 2$ is even and $i_j \neq i_{j'}$ if $j \neq j'$. Since A is structurally symmetric, there exist $k/2$ cycles in $\mathcal{G}(A, B)$, consisting of edges $\{(x_{i_1}, x_{i_2}), (x_{i_2}, x_{i_1})\}, \{(x_{i_3}, x_{i_4}), (x_{i_4}, x_{i_3})\}, \dots, \{(x_{i_{k-1}}, x_{i_k}), (x_{i_k}, x_{i_{k-1}})\}$, respectively. Those $k/2$ cycles contain k edges in total, and are disjoint with the remaining cycles and paths of $(X \cup U, \mathcal{M})$. The union of those k edges and edges of the remaining cycles and paths of $(X \cup U, \mathcal{M})$ correspond to a matching of size $|\mathcal{M}| + 1$ of $\mathcal{B}(A, B)$. We thus reach a contradiction that $\mathcal{M}$ is a maximum matching. $\square$

Proposition 1. *Suppose A is structurally symmetric. If $\mu_r(A_r, B, C) = k$, then there is a subset $X_s \subseteq X$ with $|X_s| = k$ that satisfies the following conditions: (1) X_s can be covered by a cactus configuration in $\mathcal{G}(A, B)$, and (2) X_s can be matched into Y in $\mathcal{G}(A, B, C)$.*

The proof of Proposition 1 relies on the following two intermediate results.

Lemma 5. *Let $\mathcal{B} = (V_L, V_R, E)$ be a bipartite graph. If $V_L^s \subseteq V_L$ is (left) matched by some matching of $\mathcal{B}$, then it must be matched by some maximum matching of $\mathcal{B}$.*

Proof. We prove this lemma based on the concept of augmenting path (Ahuja et al., 1993). An alternating path associated with a matching $\mathcal{M}$ is a path (i.e., sequence of edges) of the bipartite graph $\mathcal{B}$ in which edges alternate between those in $\mathcal{M}$ and those not in $\mathcal{M}$. An augmenting path w.r.t $\mathcal{M}$ is an alternating path associated with $\mathcal{M}$ that starts and ends with vertices that are

not matched by $\mathcal{M}$. Berge's theorem states that, a matching $\mathcal{M}$ is maximum if and only if there is no augmenting path w.r.t. $\mathcal{M}$ (Ahuja et al., 1993, Sec 12.6). Now, suppose $V_L^s \subseteq V_L$ is matched by a matching $\mathcal{M}_1$ which is not a maximum matching of $\mathcal{B}$. By Berge's theorem, there is an augmenting path w.r.t. $\mathcal{M}_1$ in $\mathcal{B}$, given by $\mathcal{P}_1$. It follows that the symmetric difference between $\mathcal{M}_1$ and $\mathcal{P}_1$, defined to be $\mathcal{M}_2 \doteq (\mathcal{M}_1 \cup \mathcal{P}_1) \setminus (\mathcal{M}_1 \cap \mathcal{P}_1)$, where $\mathcal{P}_1$ is regarded as a set of edges, is a new matching of $\mathcal{B}$ with $|\mathcal{M}_2| \geq |\mathcal{M}_1| + 1$. Indeed, $\mathcal{M}_2$ consists of all alternating edges of $\mathcal{P}_1$ that are not in $\mathcal{M}_1$. Since the first and last edges of $\mathcal{P}_1$ are not in $\mathcal{M}_1$, we have $|\mathcal{M}_2| \geq |\mathcal{M}_1| + 1$, and moreover, the set V_L^s is also matched by $\mathcal{M}_2$. If $\mathcal{M}_2$ is not a maximum matching, we can repeat this process, until a maximum matching $\mathcal{M}_*$ of $\mathcal{B}$ emerges that matches V_L^s . $\square$

Lemma 6. Suppose A is structurally symmetric. Let $\mathcal{M}$ be a maximum matching of $\mathcal{B}(A_r, B)$ and $X_M \subseteq X$ with $|X_M| = |\mathcal{M}|$ such that X_M is left matched by $\mathcal{M}$. Then, X_M is covered by a cactus configuration in $\mathcal{G}(A, B)$.

Proof. From Lemma 3, $(X \cup U, \mathcal{M})$ is a union of disjoint state cycles and paths that span $\mathcal{G}(A_r, B)$. Moreover, each vertex in X_M is contained in a cycle or a path with a nonzero length of $(X \cup U, \mathcal{M})$, while all paths that start from state vertices (if any) have even length. Consider one of such paths, say $P = (x_{i_1}, x_{i_2}, \dots, x_{i_k})$, where $k \geq 1$ is odd, $x_{i_1} \notin X_M$, and $i_j \neq i_{j'}$ if $j \neq j'$. By the symmetry of A , there exist $(k-1)/2$ cycles in $\mathcal{G}(A, B)$, consisting of edges $\{(x_{i_2}, x_{i_3}), (x_{i_3}, x_{i_2})\}, \{(x_{i_4}, x_{i_5}), (x_{i_5}, x_{i_4})\}, \dots, \{(x_{i_{k-1}}, x_{i_k}), (x_{i_k}, x_{i_{k-1}})\}$, respectively. Those cycles cover the state vertices $\{x_{i_2}, \dots, x_{i_k}\}$ in P that are left-matched by $\mathcal{M}$. It follows that for each of these paths, such a set of disjoint cycles can be constructed similarly. Clearly, all those newly-constructed state cycles and the remaining state cycles, as well as paths from U to X in $(X \cup U, \mathcal{M})$, are vertex-disjoint, and their union covers X_M . Notice that all input-unreachable state vertices are isolated vertices in $\mathcal{G}(A_r, B)$ as the ingoing edges to them and outgoing edges from them are removed by the definition of A_r . Therefore, X_M is covered by a cactus configuration in $\mathcal{G}(A, B)$. $\square$

Proof of Proposition 1. Suppose $\tilde{L} = (\tilde{V}_L, \tilde{E}_L)$ is a linking with the maximum size k in $\mathcal{D}_r(A_r, B, C)$. Suppose further $\tilde{V}_L \cap V_A^{(j)} = \tilde{V}_A^{(j)}$, $j = 1, 2$, $\tilde{V}_L \cap V_B^{(1)} = \tilde{V}_B^{(1)}$, and $V_L \cap V_C = \tilde{V}_C$. It is clear that there are k disjoint edges from $\tilde{V}_B^{(1)} \cup \tilde{V}_A^{(2)}$ to $\tilde{V}_A^{(1)}$. As a result, upon defining $X_s = \{x_i : x_i^1 \in \tilde{V}_A^{(1)}\}$, X_s is left matched by a matching with size k in $\mathcal{B}(A_r, B)$. From Lemma 5, X_s must also be left matched by some maximum matching of $\mathcal{B}(A_r, B)$. Lemma 6 yields that X_s is covered by a cactus configuration in $\mathcal{G}(A, B)$. Moreover, the existence of $\tilde{L}$ implies that there are k disjoint edges from $\tilde{V}_A^{(1)}$ to $\tilde{V}_C$, that is, X_s can be matched into Y . This completes the proof. $\square$

The following result extends Hosoe's characterizations for controllable subspaces of directed networks (Hosoe, 1980) to undirected networks.

Proposition 2. Let X_M be the set of state vertices covered by a cactus configuration of $\mathcal{G}(A, B)$. Let $Q(X_M, :)$ be the submatrix of Q consisting of rows corresponding to X_M , with Q defined in (4). Then, $\text{grank}_s Q(X_M, :) = |X_M|$.

Proof. The proof roughly consists of two steps. First, we construct a linking L of $\mathcal{D}(A, B)$ from a cactus configuration covering X_M . Then, we demonstrate that $\text{sign}(L)w(L)$ cannot be canceled out in $\det Q(X_M, \text{tail}(L))$.

Suppose a cactus configuration contains state cycles $C_1, \dots, C_r$ and stems $S_1, \dots, S_l$, and let $X_M \subseteq X$ be the set of state vertices

covered by it. Let us extend this cactus configuration to a subgraph $\mathcal{C}$ of $\mathcal{G}(A, B)$ such that each cycle C_i is directly connected to a unique (in $\mathcal{C}$) cycle or stem of this cactus configuration, or some input vertex of $\mathcal{G}(A, B)$, so that every vertex of C_i is input-reachable in $\mathcal{C}$ for $i = 1, \dots, r$. Moreover, if edge $(x_i, x_j) \in E_{XX}$ is in $\mathcal{C}$ then (x_j, x_i) is also included in $\mathcal{C}$. The subgraph $\mathcal{C}$ will be used to construct a linking L with size $|X_M|$ in $\mathcal{D}(A, B)$. Inspired by Poljak (1992), we describe a one-one correspondence between linkings in $\mathcal{D}(A, B)$ and 'walking schedules' in $\mathcal{G}(A, B)$. Consider a walking schedule on $\mathcal{G}(A, B)$ that satisfies the following rules: (i) At every time $t = 0, 1, \dots, n$, no two walkers can be located at the same vertex; (ii) A walker located at vertex i at time t must move into a neighboring vertex j at time $t+1$, where (i, j) is an edge of $\mathcal{G}(A, B)$; (iii) A new walker may enter $\mathcal{G}(A, B)$ only through an input vertex in U at time $t = 0, 1, \dots, n-1$.

It is clear that any schedule satisfying the above rules gives rise to a linking in $\mathcal{D}(A, B)$. Now, consider a walking schedule of $|X_M|$ walkers on $\mathcal{G}(A, B)$ that satisfies the following additional rules: (iv) Every walker uses only edges from the subgraph $\mathcal{C}$; (v) Every walker must be located at one vertex of $\mathcal{C}$ in the final time n . For each stem S_i with length $|S_i|$, the corresponding $|S_i|$ walkers enter the graph $\mathcal{C}$ at time $n - |S_i|, n - |S_i| + 1, \dots, n - 1$, respectively. For each cycle C_i , each one of the corresponding $|C_i|$ ($|\cdot|$ denotes the length) walkers walks the shortest way (in $\mathcal{C}$) to the cycle C_i . When it reaches this cycle, it continues to walk around the cycle for the remaining time units. The corresponding $|C_i|$ walkers enter $\mathcal{C}$ at time $k, k+1, \dots, k + |C_i| - 1$ for some $k \leq n - |C_i|$.

The above five rules ensure that a linking L can be constructed from the cactus configuration, with $\text{head}(L) = \tilde{X}_M \doteq \{x_i^1 : x_i \in X_M\}$. Suppose there is another linking L' with $\text{head}(L') = \text{head}(L)$, $\text{tail}(L') = \text{tail}(L)$, and $w(L') = w(L)$. Due to the vertex-disjointness of these cycles and stems and the symmetry of A , the only possible case is that for some cycles C_i , the corresponding walkers walk in the opposite directions of the original schedule associated with L in a certain period (Poljak, 1989). Consider such a cycle C_i and suppose it corresponds to a linking $L_i = (p_1, \dots, p_k)$ with $k \doteq |C_i|$. Without loss of generality, suppose $C_i = \{(x_j, x_{j+1}) : j = 1, \dots, k-1\} \cup \{(x_k, x_1)\}$ and the heads of $p_1, \dots, p_k$ are labeled $1, \dots, k$, respectively. Let $\sigma(L_i) \subseteq \{1, \dots, k\}$ be the permutation associated with L_i . In order to construct a linking L'_i associated with C_i such that $w(L'_i) = w(L_i)$, $\text{tail}(L'_i) = \text{tail}(L_i)$, and $\text{head}(L'_i) = \text{head}(L_i)$, all k walkers corresponding to L'_i , after reaching C_i , must simultaneously walk either in the direction of $x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_k \rightarrow x_1$ or the direction of $x_1 \leftarrow x_2 \leftarrow \dots \leftarrow x_k \leftarrow x_1$ (otherwise rule (i) may be violated). Let $p'_1, p'_2, \dots, p'_k$ be the resulting ordered paths in L'_i . Obviously, $\text{tail}(p'_i) = \text{tail}(p_i)$. Notice that walking in the opposite direction leads to an even number of differences in the labels of terminal vertices (see Fig. 2 for illustration). We therefore have

$$\text{head}(p'_i) = (i + 2h) \mod k, \forall i$$

for some $h \in \mathbb{N}$, where $m \mod k$ refers to the remainder obtained when m is divided by k (if m is divisible by k , the remainder is set to be k). Consequently, upon letting $\sigma(L'_i)$ be the permutation associated with L'_i , it takes an even number of transpositions to transform $\sigma(L'_i)$ into $\sigma(L_i)$, yielding $\text{sign}(L_i) = \text{sign}(L'_i)$.

The above analysis is valid for every cycle C_i in the cactus configuration. Thus, any linking L' with $\text{head}(L') = \text{head}(L)$, $\text{tail}(L') = \text{tail}(L)$, and $w(L') = w(L)$, will have the same sign as L , and hence the term $\text{sign}(L)w(L)$ cannot be canceled out in $\det Q(X_M, \text{tail}(L))$, leading to the required result.

Theorem 1. Suppose A is structurally symmetric. It holds that

$$gd_s(A, B, C) = gd(A, B, C) = \mu(A, B, C) = \mu_r(A_r, B, C).$$

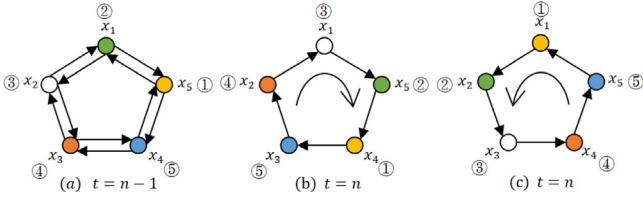


Fig. 2. Illustration of a walking schedule on a state cycle with vertices $\{x_1, \dots, x_5\}$ from time $t = n - 1$ (panel (a)) to time $t = n$ (panels (b) and (c)). Each number in the circle represents a walker, and there is a unique color to indicate its position at each time. Panel (a) depicts the positions of 5 walkers at time $t = n - 1$, while (b) and (c) depict two possible subsequent cases at time $t = n$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Proof. Based on Lemma 2, it suffices to prove that $gd_s(A, B, C) = \mu_r(A_r, B, C)$. Suppose $\mu_r(A_r, B, C) = k$. Proposition 1 yields that there is an $X_s \subseteq X$ with $|X_s| = k$ such that $X_s \subseteq X_M$, where X_M is the set of state vertices covered by a cactus configuration. By Proposition 2, rows of $Q(X_M, :)$ are generically linearly independent. Hence, the sub-rows $Q(X_s, :)$ are also generically linearly independent. Let $e_1, \dots, e_k$ be k disjoint edges from X_s to Y (via Proposition 1). Set $w(e) \equiv 0$ for the edges $e \in E_{XY} \setminus \{e_1, \dots, e_k\}$ and let C' be the resulting output matrix C . It is easy to see that $\text{grank } C'Q(X_s, :) = k$ (for example, setting $w(e_1) = \dots = w(e_k) = 1$). This means $gd_s(A, B, C) \geq \mu_r(A_r, B, C)$, leading to the required result with the fact $gd_s(A, B, C) \leq \mu_r(A_r, B, C)$ in Lemma 2. $\square$

Interestingly, it is found in Murota and Poljak (1990) that $\mu(A, B, C)$ provides an upper bound for $gd(A, B, C)$ for a general A . Meanwhile, the maximum number of state vertices that can be covered by a cactus configuration and matched into Y simultaneously provide a lower bound for $gd(A, B, C)$. The two bounds do not coincide in general. Theorem 1, however, implies that these two bounds coincide and become exact when A is structurally symmetric (either with the symmetric weight constraint or not). Based on Theorem 1, we have the following two corollaries, which express $gd_s(A, B, C)$ and $gd_s(A, B)$ in terms of the generic ranks of simple products of (at most two) matrices without the symmetry constraint. In the special case, by setting $gd_s(A, B, C) = p$ or $gd_s(A, B) = n$, these corollaries recover the known conditions for SOC in Li et al. (2021) and structural controllability in Menara et al. (2019) with the symmetry constraint.

Corollary 1. For A structurally symmetric, $gd_s(A, B, C) = gd(A, B, C) = \text{grank } C[A_r, B]$. In particular, (A, B, C) is SOC if and only if there exists $X_s \subseteq X$ with $|X_s| = p$ such that (1) $\text{grank } [A_r, B][X_s, :] = p$, and (2) X_s can be matched into Y in $\mathcal{G}(A, B, C)$.

Proof. It suffices to prove $\text{grank } C[A_r, B] = \mu_r(A_r, B, C)$. Observe that $\mathcal{D}_r(A_r, B, C)$ is exactly the dynamic graph associated with $(0_{n \times n}, [A_r, B], C)$ (the state matrix is $0_{n \times n}$). As all nonzero entries in $[A_r, B]$ and C can take independent values in computing $\text{grank } C[A_r, B]$, by letting $A = 0$ and $B = [A_r, B]$ in (2), Lemma 1 yields that $\text{grank } C[A_r, B] = \mu_r(A_r, B, C)$. $\square$

Corollary 2. For A structurally symmetric, $gd_s(A, B) = gd(A, B) = \text{grank } [A_r, B]$.

In what follows, we reveal $gd_s(A, B, C)$ and $gd(A, B, C)$ for undirected networks can be computed in polynomial time. Thus, verifying the SOC is polynomially solvable. This challenges a claim in a recent work (Li et al., 2021).

Theorem 2. If A is structurally symmetric, then both $gd_s(A, B, C)$ and $gd(A, B, C)$ can be computed in $O(n^{2.5})$ time. In particular,

verifying the SOC of (A, B, C) , whether with the symmetry constraint or not, is polynomially solvable.

Proof. Given Theorem 1, to compute $gd_s(A, B, C)$ or $gd(A, B, C)$, it suffices to compute the maximum size of a linking from $V_B^{(1)} \cup V_A^{(2)}$ to V_C in $\mathcal{D}_r(A_r, B, C)$, i.e., $\mu_r(A_r, B, C)$. This can be done by resorting to the maximum flow algorithm. Given a directed graph $\mathcal{G} = (V, E)$ with a source node s and a sink node t , where each edge (u, v) has a capacity $c(u, v)$, the maximum flow problem seeks to find the maximum flow f from s to t such that the flow on each edge does not exceed its capacity) and the conservation constraint (i.e., the total flows into a node equals the total flows out of that node, except for the source and the sink); see Ahuja et al. (1993). To determine $\mu_r(A_r, B, C)$, we can construct a flow network $\mathcal{F}(A_r, B, C)$ from $\mathcal{D}_r(A_r, B, C)$ as follows (see Fig. 4(b) for an example): For each vertex v of $\mathcal{D}_r(A_r, B, C)$, replace it with two vertices v^i, v^o and the edge (v^i, v^o) , and for each edge (v, w) of $\mathcal{D}_r(A_r, B, C)$, replace it with (v^o, w^i) . Then, add to $\mathcal{F}(A_r, B, C)$ a source s and a sink t , as well as the incident edges $\{(s, v^i) : v \in V_B^{(1)} \cup V_A^{(2)}\} \cup \{(v^o, t) : v \in V_C\}$, and set the edge capacity $c(e) = 1$ for each edge of $\mathcal{F}(A_r, B, C)$. By the integral flow theorem (that is, if each edge has an integral capacity, then there is an integral maximum flow (Ahuja et al., 1993)), it follows that the maximum flow from s to t over $\mathcal{F}(A_r, B, C)$ equals $\mu_r(A_r, B, C)$. Notice that there is no cycle in $\mathcal{F}(A_r, B, C)$ and all edge capacities are one. The maximum flow in such a flow network $\mathcal{F}(A_r, B, C)$ then can be computed in time $O(|X \cup Y \cup U|^{0.5} |E_{UX} \cup E_{XX} \cup E_{XY}|) \rightarrow O(n^{2.5})$ (Ahuja et al., 1993, Theo 8.2). Notice also that determining A_r can invoke a simple depth-first search, incurring linear complexity in $|X \cup U| + |E_{UX} \cup E_{XX}|$. Therefore, $\mu_r(A_r, B, C)$ can be obtained in time at most $O(n^{2.5})$. The remaining statements follow immediately. $\square$

Remark 1. It was reported in Li et al. (2021) that verifying the SOC of (A, B, C) with the symmetry constraint is NP-hard. Contrarily to this result, Theorem 2 reveals that this problem is indeed polynomially solvable, with or without the symmetry constraint. Li et al. (2021) provided a reduction from the three-dimensional matching (3DM) problem³ to the SOC verification problem to show its NP-hardness. More precisely, they reduced a 3DM with $T \subseteq S_1 \times S_2 \times S_3$ to a system (A, B, C) , in which links in $S_1 \times S_2$ (resp. $S_2 \times S_3$)⁴ are recast as edges in E_{UX} (resp. E_{XY}) and $E_{XX} = \emptyset$ (i.e., $A = 0$). They claimed that (A, B, C) is SOC if and only if the largest 3DM in T has a size $|S_3| (= |Y|)$. The if direction is obviously true by noting that $A = 0$. The only if direction, however, is not. This is because the existence of links (x_1, y_1) and (y_1, z_1) in $\mathcal{G}(A, B, C)$ (so in T) does not guarantee the existence of a triple (x_1, y_1, z_1) in T of a 3DM. An example illustrating this is given in Fig. 3.

Compared to the $O(n^4)$ complexity in directly computing $\mu(A, B, C)$ in the dynamic graph $\mathcal{D}(A, B, C)$ via the maximum flow algorithm, the $O(n^{2.5})$ complexity obtained via computing $\mu_r(A_r, B, C)$ significantly reduces the computational burden.

4. Minimal driver nodes for structural output controllability

In this section, we consider the problem of determining the minimum number of driver nodes (independent inputs) needed

³ Let S_1, S_2 , and S_3 be finite sets, and let T be a subset of $S_1 \times S_2 \times S_3$. $M \subseteq T$ is a 3DM if for any two distinct triples $(u_1, x_1, y_1) \in M$ and $(u_2, x_2, y_2) \in M$, $u_1 \neq u_2$, $x_1 \neq x_2$, and $y_1 \neq y_2$. The 3DM problem seeks to find the largest 3DM, which is NP-hard.

⁴ A link $(u_1, x_1) \in S_1 \times S_2$ if there is a triple $(u_1, x_1, y_1) \in T$ for some $y_1 \in S_3$; similarly for $(x_1, y_1) \in S_2 \times S_3$.

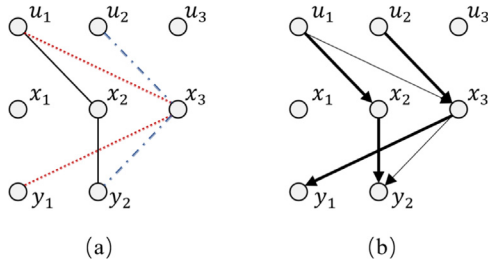


Fig. 3. (a) presents a 3DM problem with $T = \{(u_1, x_2, y_2), (u_1, x_3, y_1), (u_2, x_3, y_2)\}$, where different triples in T are depicted with distinct line styles. The corresponding system digraph $\mathcal{G}(A, B, C)$ constructed in Li et al. (2021) is given in (b). The maximum size of a 3DM in (a) is 1, while there are two disjoint paths (u_1, x_2, y_2) and (u_2, x_3, y_1) in (b), depicted in bold lines, implying the SOC of (A, B, C) via Corollary 1. Notice that although edges (u_2, x_3) and (x_3, y_1) exist in $\mathcal{G}(A, B, C)$, the triple $(u_2, x_3, y_1) \notin T$.

for SOC of undirected networks.⁵ Based on Theorem 2, we show that this problem can be solved in polynomial time. Furthermore, we show that if each driver node is *dedicated*, meaning that each driver node can directly actuate at most one state variable, corresponding to that each column of B has at most one nonzero entry, this problem becomes NP-hard.

Formally, given (A, C) with A structurally symmetric, the considered problem without constraints on B can be formulated as

finding an $n \times m$ matrix B with the minimum m ,
s.t. (A, B, C) structurally output controllable.

We call the above problem $\mathcal{P}(A, C)$. To ensure the feasibility of $\mathcal{P}(A, C)$, without losing any generality, assume that C has full row generic rank, i.e., $\text{grank } C = p$. Below, we give a closed-form solution to $\mathcal{P}(A, C)$ and a flow-network based algorithm to construct an optimal solution.

Theorem 3. Suppose A is structurally symmetric and C has full row generic rank. The optimal solution B^* to $\mathcal{P}(A, C)$ has the minimum number m^* of columns with

$$m^* = \max\{1, p - \text{grank } CA\}. \quad (5)$$

Moreover, Algorithm 1 can construct an optimal solution B^* in time $O(n^{2.5})$.

Proof. We first show that m^* is a lower bound for the number of columns of B , provided that B is feasible for $\mathcal{P}(A, C)$. Suppose that B has m columns and A_r is defined on $\mathcal{G}(A, B)$ (the definition of A_r is given in the previous section). Given that (A, B, C) is SOC, Corollary 1 yields $\text{grank } C[A_r, B] = p$. Observe that $\text{grank } C[A_r, B] \leq \text{grank } CA_r + \text{grank } CB$ and $\text{grank } CA_r \leq \text{grank } CA$. Combining these relations yield $\text{grank } CB \geq p - \text{grank } CA$. Since $m \geq \text{grank } CB$, we have $m \geq p - \text{grank } CA$. The relation $m \geq 1$ is obvious. We therefore obtain that $m \geq m^*$.

Let B^* be the matrix returned by Algorithm 1. By the construction, the number of columns of B^* is exactly m^* . Moreover, it is not difficult to see that there is a linking L with size p from $V_B^{(1)} \cup V_A^{(2)}$ to V_C in the dynamic graph $\mathcal{D}_r(A, B^*, C)$, and each path in L starts from either an input vertex or an input-reachable state vertex in $\mathcal{G}(A, B^*)$. By Theorem 1, we know that (A, B^*, C) is SOC, proving the optimality. Notice that the complexity of Algorithm 1 is dominated by the maximum flow algorithm in Step 1 and the maximum matching algorithm in Step 5. Since both algorithms have complexity $O(n^{2.5})$, it follows that the overall complexity of Algorithm 1 is $O(n^{2.5})$. $\square$

⁵ Since regardless of the symmetric weight constraint, the criteria for SOC coincide, we do not specify this constraint.

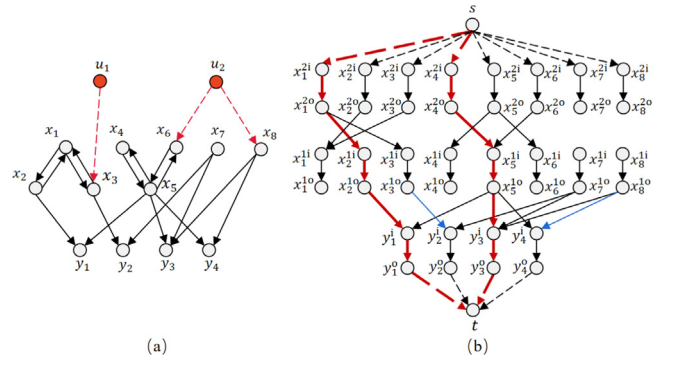


Fig. 4. Illustration of Algorithm 1. Panel (a) gives $\mathcal{G}(A, C)$ (solid lines). Panel (b) presents the corresponding flow network $\mathcal{F}(A, 0_{n \times n}, C)$, with bold red lines representing the edges of a maximum flow f^* . The blue edges correspond to a maximum matching in $(Y^c, X^c, E_{XY} \cap (X^c \times Y^c))$. In (a), the dotted lines depict an optimal solution to $\mathcal{P}(A, C)$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

When $C = I_n$, m^* given in (5) coincides with the minimal number of driver nodes needed for structural controllability (Liu et al., 2011). In view of Theorem 3, the problem of determining the minimum number of driver nodes to achieve SOC is polynomially solvable for undirected networks. By contrast, for directed networks, whether this problem can be solved in polynomial time remains open as mentioned earlier. Below, we provide an example to illustrate Algorithm 1.

Example 1. Consider a system whose $\mathcal{G}(A, C)$ is given in Fig. 4(a). The corresponding flow network $\mathcal{F}(A, 0_{n \times n}, C)$ is given in Fig. 4(b). A maximum flow from s to t in $\mathcal{F}(A, 0_{n \times n}, C)$ equals 2. This implies $\text{grank } CA = 2$ and $m^* = 2$. From this flow shown in Fig. 4(b), we get $X^f = \{x_2, x_5\}$ and $Y^f = \{y_1, y_3\}$, implying that $X^c = \{x_1, x_3, x_4, x_6, x_7, x_8\}$ and $Y^c = \{y_2, y_4\}$. A maximum matching $\mathcal{M}$ of $(Y^c, X^c, E_{XY} \cap (X^c \times Y^c))$ then can be $\mathcal{M} = \{(x_3, y_2), (x_8, y_4)\}$, which right matches x_3, x_8 . Consequently, a minimal set of 2 driver nodes is sufficient for this system to be SOC, whose structure is shown in Fig. 4(a) (not unique). It turns out that the resulting system is not structurally controllable but SOC.

In $\mathcal{P}(A, C)$, we do not impose any restrictions on the structure of B . A more practical scenario is that the number of state nodes that each driver node can actuate simultaneously is limited. A typical case is that each driver node is dedicated. We denote the problem of determining the minimal number of dedicated driver nodes needed for SOC by $\mathcal{P}^d(A, C)$. Contrarily to $\mathcal{P}(A, C)$, we show $\mathcal{P}^d(A, C)$ is NP-hard.

Theorem 4. Suppose A is structurally symmetric. The problem $\mathcal{P}^d(A, C)$ is NP-hard.

Proof. Our proof is based on a reduction from the set cover problem. Given a collection $S_1, \dots, S_r$ of subsets of $S = \{1, \dots, p\}$, the set cover problem asks the set $\mathcal{T} \subseteq \{1, \dots, r\}$ with $|\mathcal{T}|$ minimum such that $\bigcup_{i \in \mathcal{T}} S_i = S$. This problem is known to be NP-hard (Ahuja et al., 1993). Denote the i th subset by $S_i = \{i_1, \dots, i_{|S_i|}\} \subseteq \{1, \dots, p\}$. Construct (A, C) such that the associated digraph $\mathcal{G}(A, C) = (X \cup Y, E_{XX} \cup E_{XY})$ is as follows: the state vertices $X = \bigcup_{i=1}^r X_i$, with $X_i = \{x_{i1}, \dots, x_{i|S_i|}\}$, the output vertices $Y = \{y_1, \dots, y_p\}$, the edge set $E_{XX} = \bigcup_{i=1}^r E_i$, with $E_i = \{(x_{ij}, x_{ik}) : x_{ij}, x_{ik} \in X_i\}$, and $E_{XY} = \bigcup_{i=1}^r \{(x_{ij}, y_{i_j}) : j = 1, \dots, |S_i|\}$. In other words, the subgraph $\mathcal{G}(A, C) = (X, E_{XX})$ consists of r connected components $G_i = (X_i, E_i)$, $i = 1, \dots, r$, in which each $x_{ij} \in X_i$ has

a self-loop and there are bidirectional edges between any pair of vertices in X_i . Moreover, x_{ij} is connected to the i_j -th output vertex y_{ij} , $j = 1, \dots, |S_i|$. It follows that A is structurally symmetric. Let $\mathcal{T} \subseteq \{1, \dots, r\}$ be the minimal set such that $\bigcup_{i \in \mathcal{T}} S_i = S$. We claim that the optimal value of $\mathcal{P}^d(A, C)$ is $|\mathcal{T}|$.

On the one hand, consider deploying a dedicated driver node to arbitrarily one vertex in X_i , per $i \in \mathcal{T}$. Denote the resulting input matrix by B . Since every state vertex has a self-loop, $\bigcup_{i \in \mathcal{T}} X_i$ are input-reachable, and there is a subset of it that can be matched into Y in $\mathcal{G}(A, B, C)$, it follows from [Corollary 1](#) that (A, B, C) is SOC. This means the optimal value of $\mathcal{P}^d(A, C)$ is no more than $|\mathcal{T}|$. On the other hand, suppose B' corresponds to a set of dedicated driver nodes such that (A, B', C) is SOC. Let $\mathcal{T}' \subseteq \{1, \dots, r\}$ such that $\bigcup_{i \in \mathcal{T}'} X_i$ is the set of input-reachable state vertices in $\mathcal{G}(A, B')$ (since each X_i is a connected component, either all of its vertices are input reachable, or none is). By [Corollary 1](#), to make (A, B', C) SOC, there is a subset of $\bigcup_{i \in \mathcal{T}'} X_i$ that is matched into Y . This requires $|\bigcup_{i \in \mathcal{T}'} S_i| \geq |Y| = p$, leading to $|\mathcal{T}'| \geq |\mathcal{T}|$. Notice that the number of dedicated driver nodes corresponding to B' is at least $|\mathcal{T}'|$ (as $X_1, \dots, X_r$ are disjoint). As a result, the optimal value of $\mathcal{P}^d(A, C)$ is at least $|\mathcal{T}|$. We thus reach the required claim that the optimal value of $\mathcal{P}^d(A, C)$ equals $|\mathcal{T}|$. Since the set cover problem is NP-hard and the above reduction is in polynomial time, we conclude that $\mathcal{P}^d(A, C)$ is NP-hard. $\square$

In contrast to [Theorem 4](#), it has been revealed in [Pequito et al. \(2015\)](#) and [Romero and Pequito \(2018\)](#) that determining the minimum number of dedicated driver nodes required to achieve structural controllability (rather than SOC) is polynomially solvable for both directed and undirected networks. This distinction highlights the complexity associated with SOC problems. Furthermore, [Czeizler et al. \(2018, Theo 2\)](#) demonstrated that finding the minimum number of driver nodes for directed networks to achieve structural target controllability (and thus SOC) is NP-hard in practical settings, i.e., when the number of state nodes that each driver node can actuate simultaneously is bounded by a constant N , for any $N \geq 1$ (i.e., the number of nonzero entries in each column of B is bounded by N). [Theorem 4](#) extends this result to undirected networks when $N = 1$. Quite remarkably, the following corollary of [Theorem 3](#) reveals that for undirected networks, the problem of minimizing the number of dedicated driver nodes to achieve structural target controllability becomes polynomially solvable.

Corollary 3. Suppose that A is structurally symmetric and C has at most one nonzero entry in each of its rows, then the problem $\mathcal{P}^d(A, C)$ is polynomial-time solvable.

Proof. Suppose $\mathcal{G}(A)$ contains r connected components, each containing at least one target node corresponding to some nonzero columns of C . Let A_i and C_i be the state matrix and output matrix corresponding to the i -th connected component, $i = 1, \dots, r$. Since each driver node is dedicated and C has a dedicated structure, a set of driver nodes makes (A, C) SOC if and only if it makes each connected component (A_i, C_i) SOC. Therefore, it suffices to show that the minimum number m_i^* of dedicated driver nodes needed to make each connected component (A_i, C_i) SOC can be determined in polynomial time. From the proof of [Theorem 3](#), as a single driver node can make all state nodes in a connected component input-reachable, we have

$$m_i^* = \max\{1, p_i - \text{grank } C_i A_i\},$$

where p_i is the number of nonzero columns of C_i . Then, the minimum number of dedicated driver nodes required to make (A, C) SOC equals $\sum_{i=1}^r m_i^*$, which can be determined in polynomial time by [Theorem 3](#). $\square$

Algorithm 1 : Constructing an optimal solution to $\mathcal{P}(A, C)$ (A is structurally symmetric and $\text{grank } C = p$)

- 1: Find a maximum flow f^* in the flow network $\mathcal{F}(A, 0_{n \times n}, C)$ (defined in the proof of [Theorem 2](#) by setting $B = 0_{n \times n}$). Let $\mathcal{F}^*$ be the subgraph of $\mathcal{D}_r(A, 0_{n \times n}, C)$ induced by the set of vertices v such that (v^i, v^o) has a nonzero flow in f^* . Let $V_A^{(1f)}$ (resp. Y^f) be the set of vertices in $V_A^{(1)}$ (resp. V_C) covered by $\mathcal{F}^*$. Moreover, let $X^f = \{x_i \in X : x_i^1 \in V_A^{(1f)}\}$, and suppose that X^f are located at r different connected components of $\mathcal{G}(A)$, the vertex sets of which are denoted by $X_1, \dots, X_r$.
- 2: **if** $|X^f| = p$ **then**
- 3: Construct an $n \times 1$ matrix B^* as follows: $\forall j \in \{1, \dots, r\}$, arbitrarily choose one $x_k \in X_j$ and let $B_{k1}^* \neq 0$.
- 4: **else**
- 5: Find a maximum matching $\mathcal{M}$ in the bipartite graph $(Y^c, X^c, E_{XY} \cap (X^c \times Y^c))$, with $X^c = X \setminus X^f$, $Y^c = Y \setminus Y^f$. Let $X_{\mathcal{M}} = \{x_{i_1}, \dots, x_{i_m^*}\}$ be the set of right matched vertices in $\mathcal{M}$.
- 6: Construct an $n \times |\mathcal{M}|$ matrix B^* as follows: $\forall x_{i_k} \in X_{\mathcal{M}}$, let $B_{i_k, k}^* \neq 0$; $\forall j \in \{1, \dots, r\}$, if $X_{\mathcal{M}} \cap X_j = \emptyset$, arbitrarily choose one $x_k \in X_j$ and one $k' \in \{1, \dots, |\mathcal{M}|\}$, and let $B_{kk'}^* \neq 0$.
- 7: **end if**
- 8: Return B^* .

5. Structural functional observability of undirected networks

Based on our characterizations for $gd_s(A, B)$, in this section, we develop criteria for SFO of undirected networks. Consider system (1). Let $z(t) \in \mathbb{R}^r$ be the functional of states to be estimated, which is expressed as

$$z(t) = Fx(t). \quad (6)$$

Accordingly, $F \in \mathbb{R}^{r \times n}$. We denote the system described by (1) and (6) as the triple (A, C, F) .

Definition 1 (Functional Observability ([Jennings et al., 2011](#))). The system described by the triple (A, C, F) is said to be functionally observable, if for any initial state $x(0)$ and input $u(t)$, there exists a finite time T such that the value of $Fx(0)$ can be uniquely determined from the outputs $y(t)$ and inputs $u(t)$, $0 \leq t \leq T$.

It is clear that when $F = I_n$, functional observability collapses to the conventional observability ([Chen, 1984](#)). The following definition for SFO of undirected networks is a natural extension of SFO for directed networks in [Zhang et al. \(2024a\)](#).

Definition 2 (SFO). Suppose A is structurally symmetric. Let $\text{res}(A, C, F)$ be the set of triples $(\tilde{A}, \tilde{C}, \tilde{F})$ such that $\tilde{A}$ is symmetrically structurally equivalent to A and $\tilde{C}$ ($\tilde{F}$) is structurally equivalent to C (F). The triple (A, C, F) is called SFO, if almost all $(\tilde{A}, \tilde{C}, \tilde{F}) \in \text{res}(A, C, F)$ are functionally observable.

For matrices $X_1, \dots, X_k$ with compatible dimensions, $[X_1; X_2; \dots; X_k]$ denotes the matrix $[X_1^T, X_2^T, \dots, X_k^T]^T$. Let $O(A, C) = [C; CA; \dots; CA^{n-1}]$ be the observability matrix of (A, C) . From ([Jennings et al., 2011](#)) and [Zhang et al. \(2024a\)](#), a numerical triple (A, C, F) is functionally observable (c.f. [Definition 1](#)) if and only if $\text{rank } O(A, C) = \text{rank } O(A, [C; F])$. Based on this condition, the following lemma is a natural generalization of [Zhang et al. \(2024a, Prop. 2\)](#), which provides an observability matrix based criterion for SFO of undirected networks. The proof is omitted due to its similarity to [Zhang et al. \(2024a, Prop. 2\)](#).

Lemma 7. The triple (A, C, F) is SFO if and only if

$$\text{grank}_s O(A, [C; F]) = \text{grank}_s O(A, C). \quad (7)$$

By comparing Lemma 7 and the expression of $d(A, B, C)$, it is clear that the SFO of (A, C, F) is not directly dual to the SOC of (A, B, C) (that is, the SOC of (A, B, C) is not equivalent to the SFO of $(A^\top, B^\top, C^\top)$). A state vertex $x \in X$ is said to be output-reachable if there is a path from x to some output vertex $y \in Y$ in $\mathcal{G}(A, C)$. Let $X_o \subseteq X$ be the set of output-reachable state vertices in $\mathcal{G}(A, C)$. Define A_o as the matrix obtained from A after zeroing the rows and columns corresponding to $X \setminus X_o$. Define $X_F = \{x_i \in X : F_{ji} \neq 0, \exists j\}$ as the set of state vertices corresponding to the nonzero columns of F , that is, X_F is the set of state vertices that contribute directly to $z(t)$. By the duality between controllability and observability, we have $\text{grank}_s O(A, C) = \text{gd}_s(A^\top, C^\top)$. Thanks to Corollary 2, below we provide a rank-based necessary and sufficient condition for SFO of undirected networks.

Theorem 5. Suppose A is structurally symmetric. The triple (A, C, F) is SFO if and only if (1) every $x_i \in X_F$ is output-reachable, and (2) $\text{grank}[A_o; C] = \text{grank}[A_o; C; F]$.

Proof. Let $X_{o'}$ be the set of output-reachable state vertices in $\mathcal{G}(A, [C; F])$ (obtained from $\mathcal{G}(A, C)$ by regarding $[C; F]$ as the output matrix). It is clear that $X_o \subseteq X_{o'}$. Define $A_{o'}$ as the matrix obtained from A after zeroing the rows and columns corresponding to $X \setminus X_{o'}$. By Corollary 2, $\text{grank}_s O(A, C) = \text{gd}_s(A^\top, C^\top) = \text{grank}[A_o^\top, C^\top] = \text{grank}[A_o; C]$. Therefore, condition (7) is equivalent to that $\text{grank}[A_o; C] = \text{grank}[A_{o'}; C; F]$ (labeled condition $(\star)$).

Necessity: Suppose condition (1) does not hold, i.e., there is a $x_i \in X_F$ that is not output-reachable in $\mathcal{G}(A, C)$. Without loss of generality, suppose (A, C, F) is partitioned as follows

$$A = \begin{bmatrix} A_{11} & 0 \\ 0 & A_{22} \end{bmatrix}, C = [C_1, 0], F = [F_1, F_2], \quad (8)$$

where $A_{11} \in \mathbb{R}^{|X_o| \times |X_o|}$, $C_1 \in \mathbb{R}^{p \times |X_o|}$, $F_1 \in \mathbb{R}^{r \times |X_o|}$ and $F_2 \in \mathbb{R}^{r \times |X \setminus X_o|}$. In this partition, the first $|X_o|$ rows and columns of A correspond to output-reachable vertices X_o in $\mathcal{G}(A, C)$ (one can renumber the vertices to obtain such a partition). Accordingly, $A_o = \begin{bmatrix} A_{11} & 0 \\ 0 & 0 \end{bmatrix}$. By assumption, $F_2 \neq 0$. As $X_o \subseteq X_{o'}$, we get $\text{grank}[A_{o'}; C; F] \geq \text{grank}[A_o; C; F] \geq \text{grank}[A_{11}; C_1] + \text{grank} F_2 > \text{grank}[A_{11}; C_1] = \text{grank}[A_o; C]$. This implies condition $(\star)$ cannot hold. For the necessity of condition (2), observe that $\text{grank}[A_{o'}; C; F] \geq \text{grank}[A_o; C; F]$ as $X_o \subseteq X_{o'}$, and $\text{grank}[A_o; C] \leq \text{grank}[A_o; C; F]$. It is then obvious that condition $(\star)$ implies condition (2).

Sufficiency: If condition (1) holds, then $X_o = X_{o'}$. As a result, $A_o = A_{o'}$, yielding $\text{grank}[A_{o'}; C; F] = \text{grank}[A_o; C; F]$. Therefore, conditions (1) and (2) together imply condition $(\star)$, i.e., $\text{grank}[A_o; C] = \text{grank}[A_o; C; F]$. $\square$

Notice that both conditions in Theorem 5 can be verified in polynomial time. Therefore, verifying the SFO of undirected networks is polynomially solvable. Next, based on Theorem 5, we provide a graph-theoretic criterion for SFO of undirected networks. A *dilation* in $\mathcal{G}(A, C)$ is a set of state vertices $X_s \subseteq X$ satisfying $|\mathcal{N}^o(X_s)| < |X_s|$, with $\mathcal{N}^o(X_s) = \{v \in X \cup Y : (x, v) \in E_{XX} \cup E_{XY}, x \in X_s\}$ being the set of out-neighbors of X_s . A minimal dilation is a dilation such that none of its proper subsets is a dilation. Note that an isolated state vertex $x_i \in X$ without self-loops in $\mathcal{G}(A, C)$ forms a minimal dilation.

Corollary 4. For A structurally symmetric, (A, C, F) is SFO, if and only if (1) every $x_i \in X_F$ is output-reachable in $\mathcal{G}(A, C)$ and (2) no $x_i \in X_F$ is contained in a minimal dilation of $\mathcal{G}(A, C)$.

Proof. Following the proof of Zhang et al. (2024a, Prop. 3), it turns out that $\text{grank}[A_o; C] = \text{grank}[A_o; C; F]$ if and only if $\text{grank}[A_o; C] = \text{grank}[A_o; C; e_i]$ holds for each $x_i \in X_F$, where e_i is the i th row of the unity matrix I_n . Based on this fact, condition (2) of Theorem 5 holds if and only if $\text{grank}[A_o; C] = \text{grank}[A_o; C; e_i]$ for each $x_i \in X_F$. By Montanari et al. (2022, Lems 2, 5), the latter is equivalent to that, no $x_i \in X_F$ is contained in a minimal dilation of $\mathcal{G}(A_o, C)$, where $\mathcal{G}(A_o, C)$ is defined in the similar way to $\mathcal{G}(A, C)$ by changing A to A_o . It remains to prove that, this condition together with condition (1) of Theorem 5 is equivalent to conditions (1) and (2) of Corollary 4. Due to the symmetry of A , it can be seen that $\mathcal{G}(A, C)$ can be partitioned into two subgraphs, one induced by $X_o \cup Y$, given by $\mathcal{G}(A_{11}, C_1)$, and the other induced by $X \setminus X_o$, given by $\mathcal{G}(A_{22})$ (A_{11}, A_{22} , and C_1 are defined in (8)), such that there is no edge between them. As a result, there is no minimal dilation of $\mathcal{G}(A, C)$ that can intersect with both X_o and $X \setminus X_o$. Provided that $x_i \in X_F$ is output-reachable $\forall x_i \in X_F$, i.e., $X_F \subseteq X_o$, it follows that if $x_i \in X_F$ is contained in a minimal dilation of $\mathcal{G}(A_{11}, C_1)$ (so is in $\mathcal{G}(A_o, C)$), it must be also contained in a minimal dilation of $\mathcal{G}(A, C)$, and vice versa. The required result then follows. $\square$

Remark 2. From Montanari et al. (2022) and Zhang et al. (2024b), it follows that (A, C, F) is SFO with the symmetry constraint if and only if it is SFO without the symmetry constraint.

6. Conclusions

In this paper, we have characterized the generic dimensions of controllable subspace and output controllable subspace of undirected networks. We provide characterizations for them based on either the maximum linkings or generic ranks of simple matrix products. Our results show that with or without the symmetric weight constraint, these dimensions coincide. We also propose maximum flow based algorithms to verify SOC and determine the minimal driver nodes required for SOC of undirected networks in $O(n^{2.5})$ time, the former of which corrects a recent claim. We further show that the latter problem becomes NP-hard if the driver nodes are dedicated. Based on these results, we devise necessary and sufficient conditions for SFO of undirected networks. Our approach is purely graph-theoretic, based on the dynamic graphs, not relying on the eigenspace of symmetric matrices, thus providing an alternative way for analyzing the structural (output) controllability of undirected networks.

Alongside recent works on undirected networks (Belabbas & Kirkoryan, 2022; Li et al., 2021), our results suggest that certain structural properties – those determined by the zero-nonzero system structure, such as structural stability (Belabbas & Kirkoryan, 2022) and the SOC discussed here – that are challenging to characterize for directed networks may enable complete graph-theoretic characterizations and polynomial-time algorithms for undirected networks. This motivates us to further investigate additional structural properties of undirected networks, such as the structural controllability index (Mortazavian, 1982), potentially based on the dynamic graph approach.

References

- Ahuja, R. K., Magnanti, T. L., & Orlin, J. B. (1993). *Network Flows: Theory, algorithms, and applications*. Prentice Hall, Upper Saddle River, NJ.
- Belabbas, M.-A., & Kirkoryan, A. (2022). On stable systems with random structure. *SIAM Journal on Control and Optimization*, 60(1), 458–478.
- Chen, C.-T. (1984). *Linear system theory and design*. Saunders College Publishing.
- Czeizler, E., Wu, K.-C., Gratie, C., Kanhaiya, K., & Petre, I. (2018). Structural target controllability of linear networks. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 15(4), 1217–1228.

- Dion, J. M., Commault, C., & Van DerWoude, J. (2003). Generic properties and control of linear structured systems: a survey. *Automatica*, 39, 1125–1144.
- Fernando, T. L., Trinh, H. M., & Jennings, L. (2010). Functional observability and the design of minimum order linear functional observers. *IEEE Transactions on Automatic Control*, 55(5), 1268–1273.
- Gao, J., Liu, Y.-Y., D'souza, R. M., & Barabási, A.-L. (2014). Target control of complex networks. *Nature Communications*, 5(1), 1–8.
- Hosoe, S. (1980). Determination of generic dimensions of controllable subspaces and its application. *IEEE Transactions on Automatic Control*, 25(6), 1192–1196.
- Jennings, L. S., Fernando, T. L., & Trinh, H. M. (2011). Existence conditions for functional observability from an eigenspace perspective. *IEEE Transactions on Automatic Control*, 56(12), 2957–2961.
- Li, J., Chen, X., Pequito, S., Pappas, G. J., & Preciado, V. M. (2021). On the structural target controllability of undirected networks. *IEEE Transactions on Automatic Control*, 66(10), 4836–4843.
- Lin, C. T. (1974). Structural controllability. *IEEE Transactions on Automatic Control*, 19(2), 201–208.
- Liu, Y. Y., Slotine, J. J., & Barabasi, A. L. (2011). Controllability of complex networks. *Nature*, 48(7346), 167–173.
- Menara, T., Bassett, D. S., & Pasqualetti, F. (2019). Structural controllability of symmetric networks. *IEEE Transactions on Automatic Control*, 64(9), 3740–3747.
- Montanari, A. N., Duan, C., Aguirre, L. A., & Motter, A. E. (2022). Functional observability and target state estimation in large-scale networks. *Proceedings of the National Academy of Sciences (PNAS)*, 119(1), Article e2113750119.
- Mortazavian, H. (1982). On k-controllability and k-observability of linear systems. In *Analysis and optimization of systems: proceedings of the fifth international conference on analysis and optimization of systems* (pp. 600–612). Springer.
- Mousavi, S. S., Haeri, M., & Mesbahi, M. (2017). On the structural and strong structural controllability of undirected networks. *IEEE Transactions on Automatic Control*, 63(7), 2234–2241.
- Murota, K. (2009). *Matrices and matroids for systems analysis*. Springer Science Business Media.
- Murota, K., & Poljak, S. (1990). Note on a graph-theoretic criterion for structural output controllability. *IEEE Transactions on Automatic Control*, 35(8), 939–942.
- Pequito, S., Kar, S., & Aguiar, A. P. (2015). On the complexity of the constrained input selection problem for structural linear systems. *Automatica*, 62, 193–199.
- Pequito, S., Kar, S., & Aguiar, A. P. (2016). A framework for structural input/output and control configuration selection in large-scale systems. *IEEE Transactions on Automatic Control*, 61(2), 303–318.
- Poljak, S. (1989). Maximum rank of powers of a matrix of a given pattern. *Proceedings of the American Mathematical Society*, 106(4), 1137–1144.
- Poljak, S. (1990). On the generic dimension of controllable subspaces. *IEEE Transactions on Automatic Control*, 35(3), 367–369.
- Poljak, S. (1992). On the gap between the structural controllability of time-varying and time-invariant systems. *IEEE Transactions on Automatic Control*, 37(12), 1961–1965.
- Ramos, G., Aguiar, A. P., & Pequito, S. D. (2022). An overview of structural systems theory. *Automatica*, 140, Article 110229.
- Reinschke, K. J. (1988). *Multivariable control: A graph-theoretic approach*. Springer.
- Romero, O., & Pequito, S. (2018). Actuator placement for symmetric structural controllability with heterogeneous costs. *IEEE Control Systems Letters*, 2(4), 821–826.

Zhang, Y., Cheng, R., & Xia, Y. (2023). Observability blocking for functional privacy of linear dynamic networks. In *2023 62nd IEEE conference on decision and control* (pp. 7469–7474). IEEE.

Zhang, Y., Fernando, T., & Darouach, M. (2024). Functional observability, structural functional observability and optimal sensor placement. *IEEE Transactions on Automatic Control*, 1–16. <http://dx.doi.org/10.1109/TAC.2024.3462558>.

Zhang, Y., Fernando, T., & Darouach, M. (2024). Generic diagonalizability, structural functional observability and output controllability. submitted for publication.



control and analysis of network systems and data-driven analysis.



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